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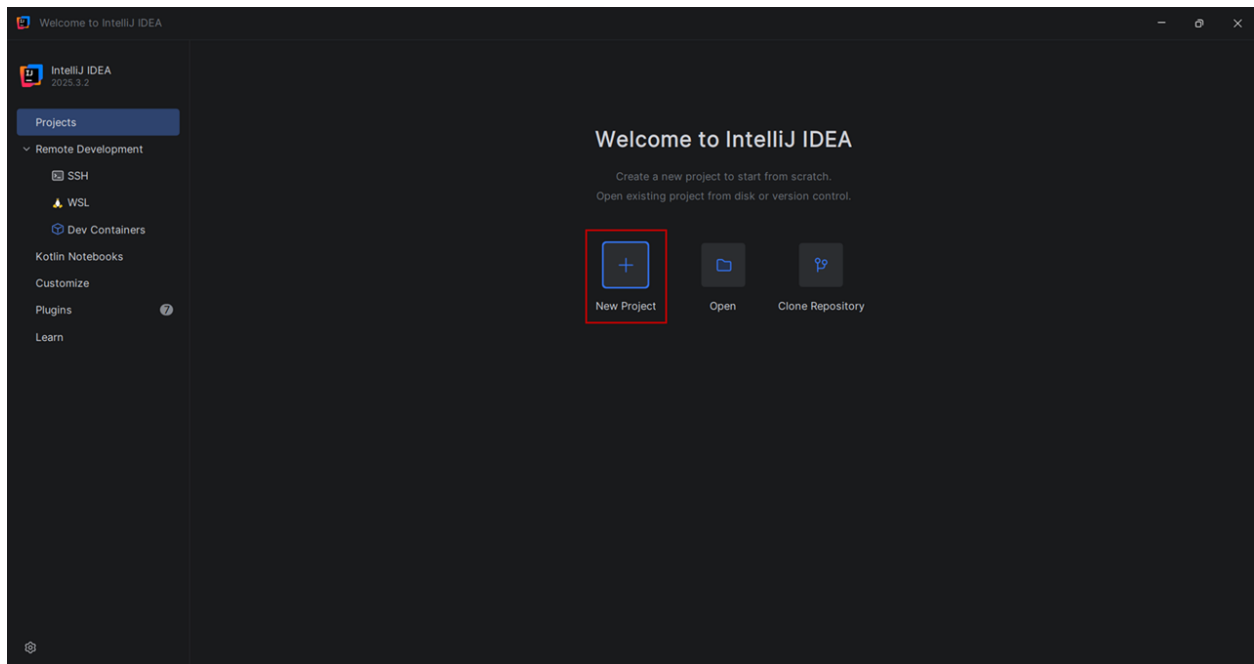
DevOps Lab Experiment – 4

Build and Run a Java Application with Maven, Migrate the Same Application to Gradle.

PART-A – Creation of Maven Project

Step 1: Create a Maven Project in IntelliJ IDEA

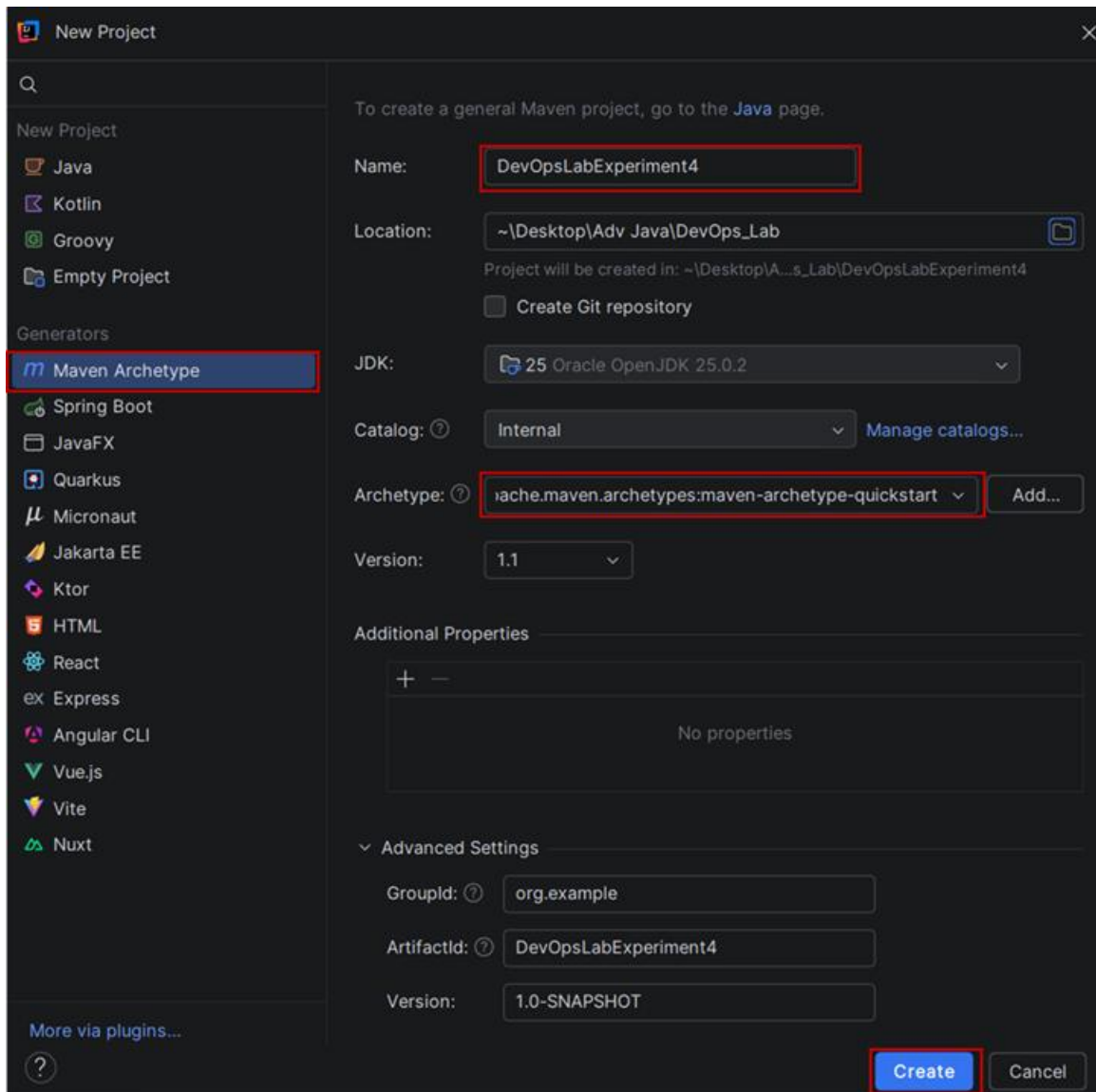
1. Open IntelliJ IDEA and click New Project



Creating a New Project

2. From the left panel, select **Maven Archetype**. This allows you to create a project using a predefined Maven template.
3. Enter the project name as **DevOpsLabExperiment4**, choose the Location where the project is to be stored and select the JDK version.
4. In the **Archetype** dropdown, select: **org.apache.maven.archetypes:maven-archetype-quickstart**. This archetype creates a simple Java project structure. Keep the Version as 1.1 (or default).
5. Expand **Advanced Settings** and enter:
GroupId: **org.example** (Represents the base package name)
ArtifactId: **DevOpsLabExperiment4** (Project name)
Version: **1.0-SNAPSHOT** (Initial development version)
6. Click **Create**. IntelliJ will generate the Maven project structure

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Configuring a Maven Project

Step 2: Update **pom.xml** to add build plugins (Maven Compiler and Jar plugins)

Paste the following **<build>** section inside the **<project>** tag in the **pom.xml** file, after the **<dependencies>** section.

- **Maven Compiler Plugin** – to compile the Java source code
- **Maven JAR Plugin** – to generate an executable JAR file by specifying the mainClass inside the manifest

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```
<build>
<plugins>

<!-- Compiler Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>

<!-- Jar Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.2.0</version>
<configuration>
<archive>
<manifest>
<mainClass>org.example.App</mainClass>
</manifest>
</archive>
</configuration>
</plugin>

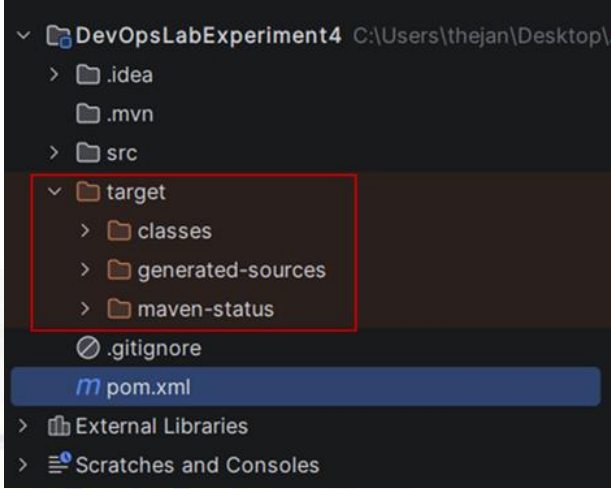
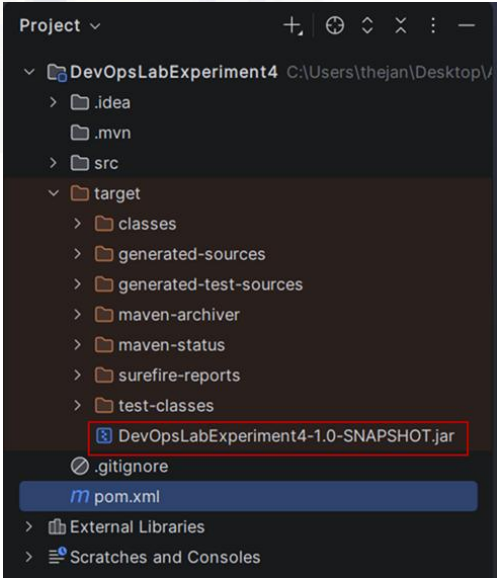
</plugins>
</build>
```

Step 3: Build and Run the Maven Project

Open the IntelliJ IDEA terminal (**press Alt + F12 to open the terminal**) & execute the following commands

```
mvn clean compile
mvn package
```

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mvn clean compile	Deletes the target/ folder (removes old build files) and compiles the source code inside src/main/java. The target is a folder automatically created by Maven when you build a project. It contains all the generated build files. 
mvn package	Compile the source code, Compile the test code, Run the tests, Create a package (usually a JAR file) inside target/ folder. 

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Step 4: Run the jar file

Locate the JAR file created inside the **target/** folder and execute it using the **java -jar** command.

java -jar target\DevOpsLabExperiment4-1.0-SNAPSHOT.jar

```
Terminal Local + -  
PS C:\Users\thejan\Desktop\Adv Java\DevOps_Lab\DevOpsLabExperiment4> java -jar target\DevOpsLabExperiment4-1.0-SNAPSHOT.jar  
Hello World!
```

PART B - Migration of Maven project to gradle project

Step 1: In the IntelliJ IDEA terminal, run the conversion command.

gradle init --type pom

This command is used to initialize a **Gradle project** from an existing **Maven project**.

Use **Groovy DSL** and the **default behavior** for the Gradle project.

```
PS C:\Users\thejan\Desktop\Adv Java\DevOps_Lab\DevOpsLabExperiment4> gradle init --type pom  
Starting a Gradle Daemon (subsequent builds will be faster)  
  
Select build script DSL:  
1: Kotlin  
2: Groovy  
Enter selection (default: Kotlin) [1..2] 2  
  
Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no] no  
  
> Task :init  
Maven to Gradle conversion is an incubating feature.  
For more information, please refer to https://docs.gradle.org/9.3.0/userguide/migrating\_from\_maven.html in the Gradle documentation.  
  
BUILD SUCCESSFUL in 1m 5s  
1 actionable task: 1 executed
```

Migrate a Maven project to a Gradle project using Groovy DSL and default settings

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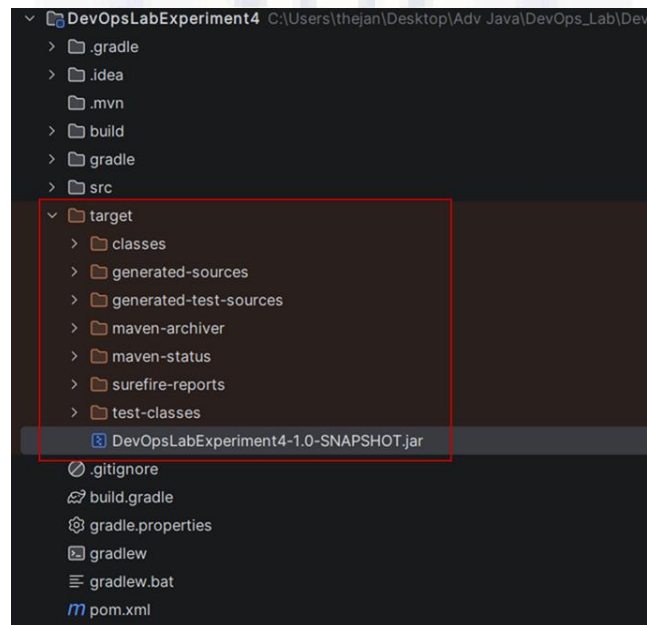
Step 2: Update **build.gradle** file by adding the below code to it and build the Gradle project using the command **gradle clean build**.

```
dependencies {  
    testImplementation 'junit:junit:4.13.2'  
}  
  
jar {  
    manifest {  
        attributes('Main-Class': 'org.example.App')  
    }  
}
```

JUnit is required because the project contains a test file (such as AppTest.java). Gradle tries to run it automatically during the project build. JUnit is not required if the AppTest.java file is deleted.

Step 3: Execute the jar file

Locate the JAR file created inside the **target/** folder and execute it using the **java -jar** command.



java -jar target/DevOpsLabExperiment4-1.0-SNAPSHOT.jar

(To get the JAR file name, type java -jar, then type the first three letters of your project name and press the **Tab** key.)

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```
PS C:\Users\thejan\Desktop\Adv Java\DevOps_Lab\DevOpsLabExperiment4> java -jar target\DevOpsLabExperiment4-1.0-SNAPSHOT.jar  
Hello World!
```

VIVA QUESTIONS

1. What is a build automation tool?
2. What is Maven and what is its purpose?
3. What is Gradle and how is it different from Maven?
4. What is the role of pom.xml in a Maven project?
5. What is the role of build.gradle in a Gradle project?
6. What is an archetype in Maven?
7. What is the purpose of maven-archetype-quickstart?
8. What are GroupId, ArtifactId, and Version in Maven?
9. What does 1.0-SNAPSHOT mean?
10. What is the target folder in Maven?
11. What does mvn clean compile do?
12. What does mvn package do?
13. How is a JAR file created in Maven?
14. How do you execute a JAR file?
15. What does “no main manifest attribute” mean?
16. Why is the Main-Class specified in the manifest file?
17. What is the purpose of gradle init --type pom?
18. How do you migrate a Maven project to a Gradle project?
19. What is Groovy DSL in Gradle?
20. Why is JUnit required in a project?